

AUTOSAR High-Level Design: Vehicle Control Unit (VCU)

Electric Vehicle Powertrain Control | Revision: 2.0

Document ID	SYS-HLD-VCU-REV-2.0	Target ECU	ECU_VCU_TC397
AUTOSAR Release	Classic Platform 4.4.0	Safety Level	ISO 26262 ASIL-D
Subsystem	EV Powertrain & Torque Mgmt	Revision	Version 2.0

1. System Overview and Torque Coordination

The Vehicle Control Unit (VCU) serves as the master supervisory controller for an electric vehicle powertrain. In revision 2.0, the architecture implements multi-core AUTOSAR Classic mapping on Infineon AURIX TC397. The VCU performs driver pedal interpretation, traction torque calculation, regenerative braking blending, and battery state-of-charge (SoC) power derating.

2. SWC Architecture & Allocation

SWC Name	Type	Periodicity	Core Allocation	Role
SWC_TorqueCoordinator	ApplicationSWC	5 ms	Core 0 (Lockstep)	High-frequency torque control with ASIL-D plausibility check.
SWC_RegenBrakeManager	ApplicationSWC	10 ms	Core 0	Blends hydraulic friction and electric motor regenerative deceleration.
SWC_ThermalSupervisor	ApplicationSWC	50 ms	Core 1	Upgraded multi-zone thermal management for fast-charging support.
SWC_BatteryStateMonitor	ApplicationSWC	20 ms	Core 2	NEW in v2.0: Active cell balancing coordination and high-voltage isolation fault detector.

3. Inter-SWC and Inverter Interfaces

Communication with the Traction Inverter occurs over an isolated CAN FD private bus (5 Mbps baud rate) via AUTOSAR COM and CanTp. Torque requests are governed by interface *If_MotorTorqueCommand* with a freshness counter and end-to-end (E2E) Profile 4 protection.